Test-Navigation

1 2 3 4 5 6 7

Seiten einzeln anzeigen

Überprüfung beenden

Begonnen am	Tuesday, 30. June 2020, 17:30
Status	Beendet
Beendet am	Tuesday, 30. June 2020, 19:00
Verbrauchte Zeit	1 Stunde 29 Minuten
Bewertung	**91,00** von 100,00

Frage **1**
Vollständig
Erreichte Punkte 10,00 von 12,00
Frage markieren

Example 1a-c)

- a) Explain with a few words, the difference between a quantitative-discrete attribute and a quantitative-continuous attribute in descriptive statistics. State one example for each kind of these attributes. [4 points]
- b) Explain with a few words, what is meant by a "type I error" and a "type II error" in inductive statistics. [4 points]
- c) Explain with a few words, what is meant by "Central Limit Theorem" in statistics. [4 points]

a) You can count quantitative-discrete attributes like the number of apples on trees.

Quantitative-continuous attributes are not countable in this sense. You can measure this attributes with an infinite precision. For example when you have the attribute body height. You can measure it in meters, in centimeters, in millimeters and so on.

b) Type I error means that an experiment finds a difference but in the population (in real) there is no difference.

Type II error means that the experiment finds no difference but in the population is a difference.

c) When your sample is big enough (> 30) then it converges to a normal distribution. The bigger it is the more it looks like a normal distribution.

Kommentar:
c) sum converges to normal distribution (-2 points)

Frage **2**
Teilweise richtig
Erreichte Punkte 3,00 von 4,00
Frage markieren

Example 1d)

Mark only the correct statements. None, one or more of the 4 given statements may be correct. [4 points, no minus points for wrong answers]

Wählen Sie eine oder mehrere Antworten:

- [ ] The Bravais-Pearson correlation coefficient has a range from $-\infty$ to $+\infty$.
- [ ] The statement $P(A \cap B) = P(A) * P(B)$ holds for any event without any restrictions.
- [x] Given a continuous random variable, the probability density function f(x) is exactly the probability that the random variable is x. ✗
- [x] The variance has a range from 0 to $+\infty$. ✓

Die Antwort ist teilweise richtig.

Sie haben zu viele Optionen gewählt.
Die richtige Antwort ist: The variance has a range from 0 to $+\infty$.

Frage **3**
Vollständig
Erreichte Punkte 10,00 von 10,00
Frage markieren

Example 2)

Three children, Ann, Berta and Chris take turns doing the dishes each night of the week (= 7 days). At the beginning of each week they make a schedule. If Ann does the dishes on three days, and Berta and Chris each do them twice, how many different schedules are possible? [10 points]

210 possibilities

permutation with repetition

7!/(3!*2!*2!) = 210

Frage **4**
Vollständig
Erreichte Punkte 9,00 von 15,00
Frage markieren

Example 3a-b)

During a bike tour of 7 days, the travelled distances for each day (rounded to whole kilometers) were recorded: 90 ; 71 ; 75 ; 80 ; 70 ; 85 ; 81

- a) Calculate the average distance travelled and the median distance travelled. [7 points]
- b) Calculate the variance and standard deviation of this population. [8 points]

a) x_bar = 78.857 (kilometers/day) = (70+71+75+80+81+85+90)/7

arithmetic mean used

median = 80

order the values and use the value in the middle (the fourth value). This is the median.

b) sigma² = 53.044 = (1/6) * ((70 - x_bar)² + (71-x_bar)²+(75-x_bar)²+(80-x_bar)²+(81-x_bar)²+(85-x_bar)²+(90-x_bar)²)

sigma = 7.283 = square_root(sigma²)

Kommentar:
b) V(X) = 46.122, SD(X) = 6.79 (-6 points)

Frage 5
Vollständig
Erreichte Punkte 20,00 von 20,00
Frage markieren

Example 4a-b)

Among the participants of a conference 60% are living in the USA. 1 out of 8 of the participants who are living in the USA are eating pancakes for breakfast. Only 1 out of 80 of the participants who are not living in the USA are eating pancakes for breakfast.

- a) What percentage of the participants are eating pancakes for breakfast? [10 points]
- b) How likely is it, that a participant is living in the USA, given it is observed that this participant is eating pancakes for breakfast? [10 points]

a) P(pancake) = 8.000 % = P(pancake and USA) + P(pancake and not_USA)

P(pancake and USA) = 7.500 %

P(pancake and not_USA) = 0.500 %

P(pancake| USA) = 1/8 = P(pancake and USA)/P(USA)

P(USA) = 0.6

b) P(USA|pancake) = 93.750 % = P(Pancake and USA)/P(pancake) = 0.075/0.08

Frage 6
Vollständig
Erreichte Punkte 18,00 von 18,00
Frage markieren

Example 5a-b)

On behalf of the owner of a winery, the true average bottled quantity of wine, which is bottled in 750 ml wine bottles, should be estimated based on a 95% confidence interval. The filling quantity X is regarded as normally distributed random variable. 10 bottles are randomly selected, and the filling quantities of these bottles are checked and the following results were calculated based on this sample.

$$\overline{x} = 749.8ml \qquad s = 5.07ml$$

- a) Calculate the 95% confidence interval. Interpret the confidence interval with a few words. [9 points]
- b) The owner of this winery now wants to determine a confidence interval for the average filling quantity, which is 2 units long and has still the same reliability of 95%. Calculate the approximately needed sample size for this confidence interval. The length of a confidence interval is defined from its lower limit to its upper limit. [9 points]

a) [746.177; 753.423]

sigma unknown --> t-test

t_(9.0.975) = 2.26

749.8 + or - 2.26 * (5.07/square_root(10))

mu (the mean of a population) is with a probability of 95 % between a value of 746.177 and 753.423. So the true average wine in one bottle lays between 746.177 and 753.423 ml.

b) n shall be at least 99.

u_0.975 = 1.96

n circa= (2* 1.96*(5.07/2))²

Frage 7
Vollständig
Erreichte Punkte 21,00 von 21,00
Frage markieren

Example 6a-b)

In a psychological experiment the research question is whether the average number of points in an empathy test of a sample of 21 persons with a mean of 82.5 points and a standard deviation of 6.2 points differs statistically significantly from the reference value of a specific population (80 points) or not.

- a) Formulate appropriate statistical hypotheses and perform an appropriate statistical test to answer this research question. This statistical test should have a type I error of α = 5%. The data can be considered to be normally distribution. State the test statistic, the critical value and the decision of your test. [14 points]
- b) Assuming the statistical test now has a type I error of α = 10%. What are the test statistic, the critical value and the test decision now? [7 points]

a) H_0: mu_sample = mu_population   sample not differs from population

H_1: mu_sample != mu_population   sample differs from population

T(x) = 1.848 = ((82.5 - 80)/6.2)*square_root(20)

t_(20.0.975) = 2.09

|T(x)| < t_(20.0.975)

We cannot reject H_0. That means we cannot make any statements from the sample to the population.

b) T(x) = 1.848 (same calculation again)

t_(20.0.95) = 1.73

|T(x)| > t_(20.0.95)

Now we can reject H_0. That means that we can assume that the sample mean differs from the population mean with a error probability of 10 %.